

Submission Notes

A2 / R-157 / R-158

Prepared for Jusang Lee

Package v0.1.42 – 9 September 2026

Review status: external review not performed. The submission materials are prepared at the author's request without waiting for external review. Preparation is not scientific acceptance, a priority certificate, or an actual journal submission.

1. What is being submitted

Global Well-Posedness, Neutral-State Rejection, and Ensemble-Induced Shell Coexistence in a Regularized Multicomponent Brazovskii Functional.

The paper concerns one explicitly printed regularized classical functional on the fixed side-16 three-torus, with three complex (six real) field components. The positive density floor, coefficients, pairing, Laplacian convention and function spaces are part of the theorem, not optional implementation choices.

The proposed contribution is the conjunction of:

1. Global well-posedness for all H^2 data, energy dissipation and positive-time smoothness of the displayed gradient flow.
2. Exact neutral coercivity and radial rejection, unique zero critical point/global minimizer, and exponential decay.
3. A separately imposed charge/chemical-potential ensemble with explicit finite-shell saturation and zero/nonzero coexistence.

Version 0.1.42 retains the v0.1.41 mathematical body and changes only publication-status/disclosure prose and submission materials.

2. What the paper does not claim

No physical vacuum, BCC selection, derived physical charge, real-time quantum dynamics, quantum or infinite-volume limit, cosmology or Sector-A closure is claimed. The registered conditional tiers and the unresolved canonical-source transfer issue are unchanged. R-472 is non-bearing assurance only. The paper does not claim the standard analytic methods are individually new.

3. Review and computational assistance

No external mathematical or signed specialist novelty review has been obtained. The blank proof and novelty forms are invitations to criticism and contain no reviewer outcome. Their hashes identify the reviewed object, not an endorsement.

AI-assisted tools supported research discussion, drafting and internal criticism of proof text, verification-code development and document preparation. They are neither authors nor independent reviewers. The named author remains responsible for mathematical correctness, references and the final submission. The literature crosswalk is bounded internal work, not exhaustive novelty certification. A counterexample or subsuming result requires reassessment.

4. File routing and reproducibility

Main manuscript

`manuscript.pdf` and `manuscript.tex`. The bibliography is included in the source.

Editor material

`cover-letter.md` and these submission notes.

Scientific supplement

The proof audit, theorem-applicability audit, literature crosswalk, internal re-review and canonical-transfer explanation.

Verification

The verification README, scripts, exact inputs, run JSON and reproduction manifest. Finite checks support auditability; they do not establish the analytic proof by execution.

Transport identity

The package builder records every archived file and its SHA-256. A fresh extraction must reproduce the declared checks. The source snapshot is distinct from Python, Lean and TeX runtime dependencies.

Install only the dependencies declared by the reproduction protocol. The historical full suite requires its pinned SymPy, Lean/Mathlib and Tectonic inputs; a resolved external cache is an environment input and must be reported. No private cache is presented as included source. Current receipts identify which checks were actually executed and which PDF was inspected.

5. Before the author presses Submit

1. Select a journal and check its current format, source, supplementary material, code/data and AI-use requirements.
2. Confirm the author identity and contact, affiliation, funding, competing interests, contributions, prior dissemination and any exclusivity attestation. No answer to these personal questions is fabricated here.
3. Approve the final manuscript and cover letter. Freeze the actual integrated commit and separately authorize the required submission tag.
4. Upload the approved files and retain the real submission receipt. Only then record the lifecycle as submitted. Acceptance/publication requires its own real event.

These author and venue actions are not a requirement to find an external pre-submission reviewer. No claim-level operator-confirmed capstone grade, journal acceptance or submission authorization is inferred from package QA.